1. Let's make a backup user account so that the root account is safe.

```
[root@localhost ~]# cat /etc/shadow
root:$6$bdYcfa2D3zf/QcyY$Dmj9UgYDy5tFVnjeBLhk3Dvzey0jwfnBYugmC.Gp4ijDRMFMe2syvkUucxD/5jZ1.WyY231DaUeILAFf4KnnN/::0:99999:7:::
bin:*:17110:0:99999:7:::
daemon:*:17110:0:99999:7:::
adm:*:17110:0:99999:7:::
lp:*:17110:0:99999:7:::
sync:*:17110:0:99999:7:::
shutdown:*:17110:0:99999:7:::
halt:*:17110:0:99999:7:::
mail:*:17110:0:99999:7:::
operator:*:17110:0:99999:7:::
games:*:17110:0:99999:7:::
ftp:*:17110:0:99999:7:::
nobody:*:17110:0:99999:7:::
systemd-network:!!:17630::::::
dbus:!!:17630::::::
polkitd:!!:17630::::::
postfix:!!:17630::::::
chrony:!!:17630::::::
sshd:!!:17630::::::
med:!!:18217::::::
ckups:!!:19822:0:99999:7:::
oot@localhost ~]# _
```

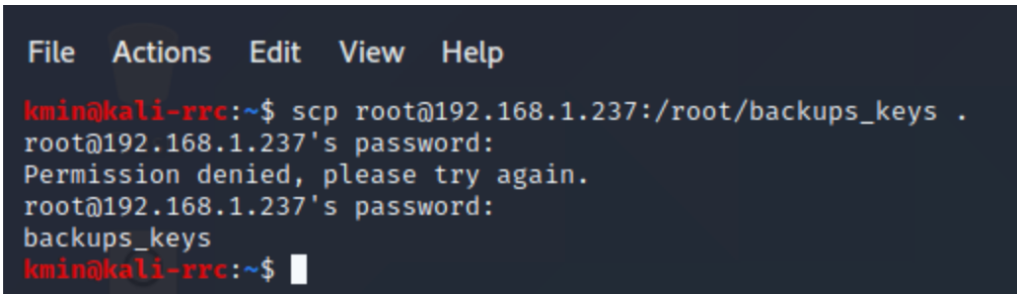
2. If you want to backup automatically, you may need to set up ssh so that the automation process doesn't have to include the password. Let's generate the public and private keys to set up ssh.

```
[root@localhost ~]# ssh-keygen -t ecdsa -f backups_keys
Generating public/private ecdsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in backups_keys.
Your public key has been saved in backups_keys.pub.
The key fingerprint is:
SHA256:ecZpleboAIlh7JHJhaChH04Y+G0vZkOXzEmfBvedsIY root@localhost.localdomain
The key's randomart image is:
+---[ECDSA 256]---+
|X.=.             |
|=& .  o . .      |
|*.*. + *o+.+ .   |
| =.o+ **E++oo    |
|  oo.o..S.*      |
|    =o.  +       |
|   o o.          |
|                 |
|                 |
+----[SHA256]-----+
oot@localhost ~]# ls
aconda-ks.cfg  backups  backups_keys  backups_keys.pub  shadow_output.txt
oot@localhost ~]#
```

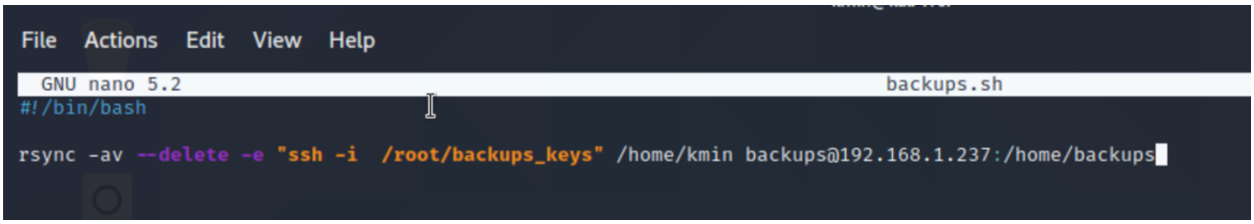
3. We need to set up the ssh as always.

```
[root@localhost ~]# mkdir /home/backups/.ssh
[root@localhost ~]# ls /home/backups/
[root@localhost ~]# ls -a /home/backups/
.  ..  .bash_logout  .bash_profile  .bashrc  .ssh
[root@localhost ~]# cp backups_keys.pub /home/backups/.ssh/authorized_keys
[root@localhost ~]# ls -l /home/backups/.ssh
total 4
-rw-r--r--. 1 root root 188 Apr  9 11:34 authorized_keys
[root@localhost ~]# ls -la /home/backups/
total 12
drwx------. 3 backups backups  74 Apr  9 11:33 .
drwxr-xr-x. 3 root    root     21 Apr  9 11:26 ..
-rw-r--r--. 1 backups backups  18 Nov 24  2021 .bash_logout
-rw-r--r--. 1 backups backups 193 Nov 24  2021 .bash_profile
-rw-r--r--. 1 backups backups 231 Nov 24  2021 .bashrc
drwxr-xr-x. 2 root    root     29 Apr  9 11:34 .ssh
[root@localhost ~]# _
```

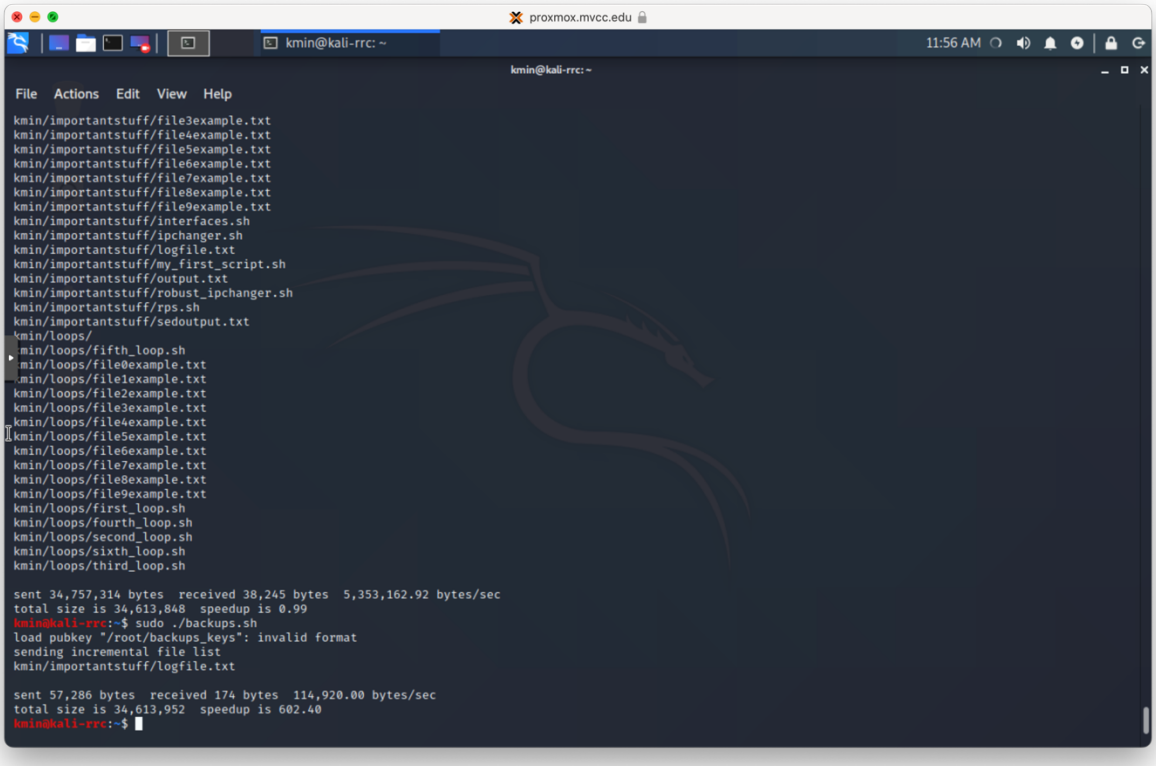
4. Copy the private keys to the user side.



5. Let's make a script that backs up the computer using ssh passwordless authentication. To use the passwordless authentication, you need to put the whole ssh -i command in the quote.



6. If you run the script, your data will start backing up using ssh key authorization.



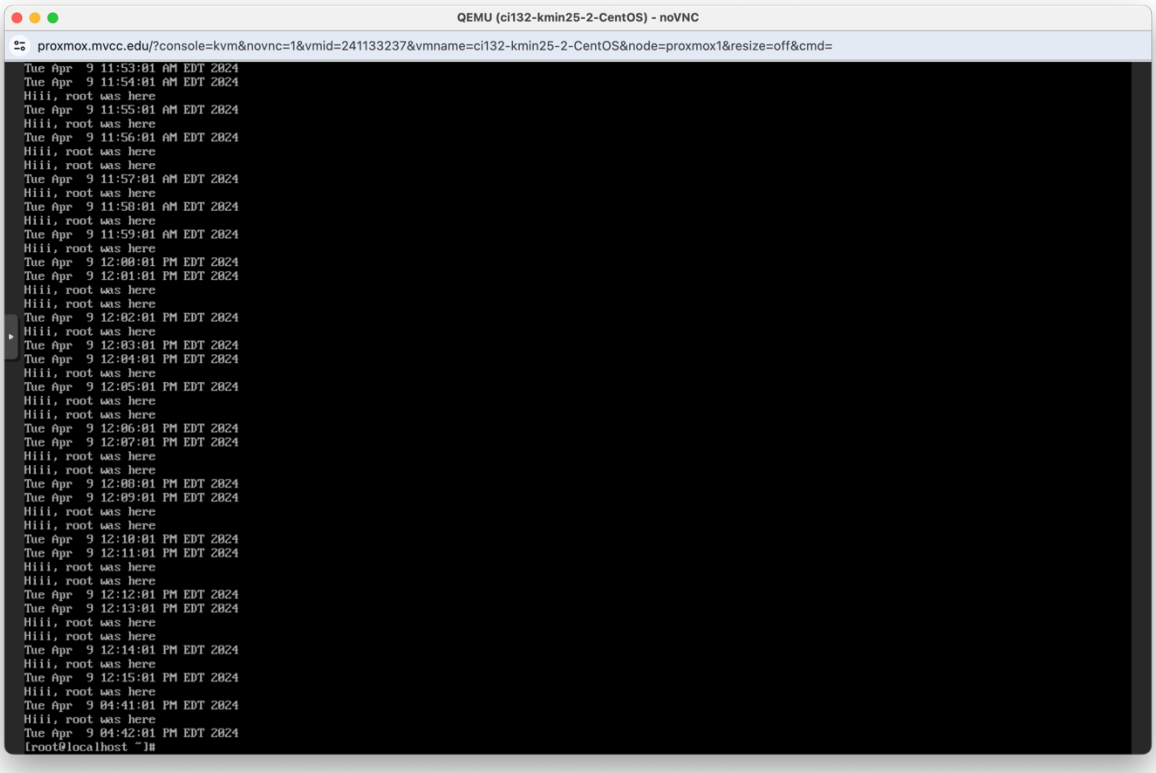
7. Now, you may need to run the script within crontab so that the backup will be automatic.

```
  GNU nano 5.2                                                    /etc/crontab
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab'
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

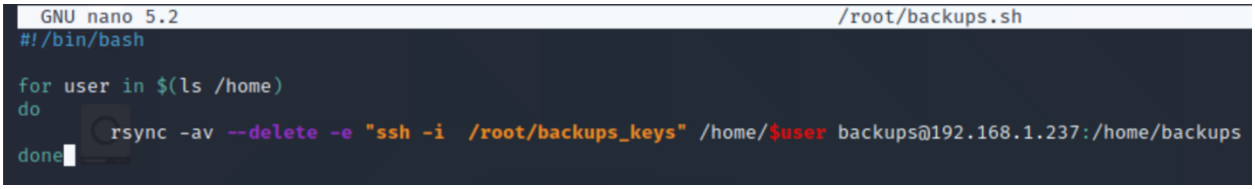
SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/sbin:/bin:/usr/sbin:/usr/bin

# Example of job definition:
# .---------------- minute (0 - 59)
# |  .------------- hour (0 - 23)
# |  |  .---------- day of month (1 - 31)
# |  |  |  .------- month (1 - 12) OR jan,feb,mar,apr ...
# |  |  |  |  .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# |  |  |  |  |
# *  *  *  *  * user-name command to be executed
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily )
47 6    * * 7   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.weekly )
52 6    1 * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.monthly )
#
* * * * * root /home/kmin/root_cronjob.sh
* * * * * root /root/backups.sh
```

8. Here's the result of running the backup automatically.



9. If you want to make backups for all of your users, you can make a loop.



10. If you restart your service, you now will have a new user file in your backup server.

